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Connection with
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process

FLT for decoupled renewal
processes

LSL and LIL for decoupled
renewal processes

Application

Limit theorems for decoupled renewal processes

Iryna Feshchenko, Kyiv, Ukraine

based on a joint paper with C. Dong and A. Iksanov

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$\xi_1, \xi_2, \dots$ – independent copies of $\xi > 0$.

$S_n = \xi_1 + \xi_2 + \dots + \xi_n$ for $n \in \mathbb{N}$, $(S_n)_{n \geq 1}$ – *standard random walk*

$N(t) := \sum_{n \geq 1} \mathbb{1}_{\{S_n \leq t\}}$ for $t \geq 0$, $(N(t))_{t \geq 0}$ – *renewal process*



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Let $\hat{S}_1, \hat{S}_2, \dots$ be independent random variables such that, for each $n \in \mathbb{N}$, $\hat{S}_n$ has the same distribution as S_n .

$(\hat{S}_n)_{n \geq 1}$ – *decoupled random walk*

$\hat{N}(t) := \sum_{n \geq 1} \mathbb{1}_{\{\hat{S}_n \leq t\}}$ for $t \geq 0$, $(\hat{N}(t))_{t \geq 0}$ – *decoupled renewal process*



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Let $\mathbb{C}$ denote the set of complex numbers. For $\rho > 0$, define the kernel C_ρ by

$$C_\rho(z, w) = \frac{\rho}{2\pi} E_{2/\rho, 2/\rho}(z\bar{w}) e^{-\frac{|z|^\rho}{2} - \frac{|w|^\rho}{2}}, \quad z, w \in \mathbb{C},$$

where $E_{a,b}$ denotes Mittag-Leffler function with parameters a and b :

$$E_{a,b}(z) := \sum_{k \geq 0} \frac{z^k}{\Gamma(ak + b)}, \quad z \in \mathbb{C},$$

and Γ is the Euler gamma-function.



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Denote by Θ_ρ a simple point process on $\mathbb{C}$ such that, for any $k \in \mathbb{N}$ and any mutually disjoint Borel subsets $B_1, \dots, B_k$ of $\mathbb{C}$,

$$\mathbb{E} \prod_{j=1}^k \Theta_\rho(B_j) = \int_{B_1 \times \dots \times B_k} \det(C(z_i, z_j))_{1 \leq i, j \leq k} dz_1 \dots dz_k.$$

The point process Θ_ρ is a *determinantal point process* with kernel C_ρ with respect to Lebesgue measure on $\mathbb{C}$.

The process by Θ_2 is known as the *infinite Ginibre point process*.



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Kostlan (1992) result: the set of absolute values of the atoms of Θ_2 has the same distribution as $(\hat{S}_n^{1/2})_{n \in \mathbb{N}}$, where $(\hat{S}_n)_{n \in \mathbb{N}}$ is a decoupled random walk with $\hat{S}_1 \sim \text{Exp}(1)$.

Fenzl and Lambert (2022) proved a FLT for $(\Theta_2(D_t))_{t \geq 0}$, properly scaled, normalized and centered, where $D_t := \{z \in \mathbb{C} : |z| < t^{1/2}\}$.

Kostlan result entails $(\Theta_2(D_t))_{t \geq 0}$ has the same distribution as $\left(\sum_{n \geq 1} \mathbb{1}_{\{\hat{S}_n \leq t\}}\right)_{t \geq 0} = (\hat{N}(t))_{t \geq 0}$, where $\hat{S}_1 \sim \text{Exp}(1)$.



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Adhikari (2018) result: the set of absolute values of the atoms of Θ_ρ has the same distribution as $(\hat{S}_n^{1/\rho})_{n \in \mathbb{N}}$, where $\hat{S}_1$ has the gamma distribution with parameters $2/\rho$ and 1.



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Put $\tau(t) := \inf\{k \geq 1 : S_k > t\}$ for $t \geq 0$.

$$\frac{\tau(t) - \mu^{-1}t}{\mu^{-1-1/\alpha}c_\alpha(t)} \xrightarrow{d} Z_\alpha, \quad t \rightarrow \infty,$$

where $\mu := \mathbb{E}\xi < \infty$. In what follows ℓ denotes a function slowly varying at ∞ .

(A) $\sigma^2 = \text{Var } \xi \in (0, \infty)$; $\alpha = 2$, $c_2(t) = \sigma t^{1/2}$;

(B) $\sigma^2 = \infty$ and $\mathbb{E}[\xi^2 \mathbb{1}_{\{\xi \leq t\}}] \sim \ell(t)$ as $t \rightarrow \infty$; $\alpha = 2$ and

$$\lim_{t \rightarrow \infty} \frac{t\ell(c_2(t))}{(c_2(t))^2} = 1.$$

(C) $\mathbb{P}\{\xi > t\} \sim t^{-\alpha}\ell(t)$ as $t \rightarrow \infty$ for some $\alpha \in (1, 2)$ and

$$\lim_{t \rightarrow \infty} \frac{t\ell(c_\alpha(t))}{(c_\alpha(t))^\alpha} = 1.$$



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(B) $\sigma^2 = \infty$ and $\mathbb{E}[\xi^2 \mathbb{1}_{\{\xi \leq t\}}] \sim \ell(t)$ as $t \rightarrow \infty; \alpha = 2;$

(C) $\mathbb{P}\{\xi > t\} \sim t^{-\alpha} \ell(t)$ as $t \rightarrow \infty$ for some $\alpha \in (1, 2).$

$$\mathbb{P}\{\xi > t\} \tau(t) \xrightarrow{d} Z_\alpha, \quad t \rightarrow \infty$$

(D) $\mathbb{P}\{\xi > t\} \sim t^{-\alpha} \ell(t)$ as $t \rightarrow \infty$ for some $\alpha \in [0, 1)$, Z_α has Mittag-Leffler distribution with parameter α characterized by its moments

$$\mathbb{E}[(Z_\alpha)^n] = \frac{n!}{(\Gamma(1-\alpha))^n \Gamma(1+n\alpha)}, \quad n \in \mathbb{N}.$$



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$$\mathbb{E}[(Z_\alpha)^n] = \frac{n!}{(\Gamma(1 - \alpha))^n \Gamma(1 + n\alpha)}, \quad n \in \mathbb{N}.$$

[Alsmeyer, Iksanov & Kabluchko \(2025\)](#) result: FCLTs for regimes (A), (B) and (C).



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Put $G(t) := 1/\mathbb{P}\{\xi > t\}$ for $t \geq 0$. For $\alpha \in [0, 1)$, denote by h_α any nonnegative nondecreasing function satisfying $G(h_\alpha(t)) \sim e^t$ as $t \rightarrow \infty$.

Theorem 1 (Dong, F. & Iksanov (2026))

Suppose (D). Then

$$\left(\frac{\hat{N}(h_\alpha(t+u)) - \mathbb{E}\hat{N}(h_\alpha(t+u))}{e^{t/2}} \right)_{u \in \mathbb{R}} \Longrightarrow X_\alpha, \quad t \rightarrow \infty$$

on $D(\mathbb{R})$, where $X_\alpha = (X_\alpha(u))_{u \in \mathbb{R}}$ is a centered Gaussian process with covariance function

$$\begin{aligned} \text{Cov}(X_\alpha(u), X_\alpha(v)) \\ = \int_0^\infty \mathbb{P}\{Z_\alpha > e^{-(u \wedge v)} y\} \mathbb{P}\{Z_\alpha \leq e^{-(u \vee v)} y\} dy \end{aligned}$$

for $u, v \in \mathbb{R}$.



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Lemma 1

Suppose (D). Then

$$\begin{aligned} \lim_{t \rightarrow \infty} \frac{\text{Cov}(\hat{N}(h_\alpha(t+u)), \hat{N}(h_\alpha(t+v)))}{e^t} \\ = \int_0^\infty \mathbb{P}\{Z_\alpha > e^{-(u \wedge v)} y\} \mathbb{P}\{Z_\alpha \leq e^{-(u \vee v)} y\} dy \end{aligned}$$

for $u, v \in \mathbb{R}$.



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Put

$$Z(t, u) := e^{-t/2} (\hat{N}(h_\alpha(t+u)) - \mathbb{E}\hat{N}(h_\alpha(t+u))), \quad u \in \mathbb{R}.$$

Step 1: finite-dimensional convergence

- Apply the Lindeberg–Feller theorem.
- Convergence of second moments follows from Lemma 1.

Step 2: tightness

- Show that for all $u < v < w$ in the interval $[-A, A]$, large $t > 0$ and a constant $C > 0$

$$\mathbb{E}[(Z(t, v) - Z(t, u))^2 (Z(t, w) - Z(t, v))^2] \leq C(e^w - e^u)^2.$$



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Theorem 2 (Dong, F. & Iksanov (2026))

Suppose (A). Then

$$\limsup_{t \rightarrow \infty} \frac{\hat{N}(t) - \mathbb{E}\hat{N}(t)}{t^{1/4}(\log t)^{1/2}} = \left(\frac{\sigma^2}{\mu^3 \pi} \right)^{1/4} \text{ a. s.},$$

where $\mu = \mathbb{E}\xi < \infty$.

Suppose (B). Then

$$\limsup_{t \rightarrow \infty} \frac{\hat{N}(t) - \mathbb{E}\hat{N}(t)}{(c_2(t) \log t)^{1/2}} = \left(\frac{1}{\mu^3 \pi} \right)^{1/4} \text{ a. s.}$$



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Suppose (C). Then

$$\begin{aligned} & \limsup_{t \rightarrow \infty} \frac{\hat{N}(t) - \mathbb{E}\hat{N}(t)}{(c_\alpha(t) \log t)^{1/2}} \\ &= \left(\frac{2(\alpha - 1)\Gamma(1 - 1/\alpha)}{\alpha\mu^{1+1/\alpha}\pi} \left(\frac{-2\Gamma(2 - \alpha) \cos(\pi\alpha/2)}{\alpha - 1} \right)^{1/\alpha} \right)^{1/2} \text{ a. s.,} \end{aligned}$$

where Γ is the Euler gamma-function.

Suppose (D). Then

$$\begin{aligned} & \limsup_{t \rightarrow \infty} \left(\frac{\mathbb{P}\{\xi > t\}}{\log \log t} \right)^{1/2} (\hat{N}(t) - \mathbb{E}\hat{N}(t)) \\ &= \left(2 \int_0^\infty \mathbb{P}\{Z_\alpha > y\} \mathbb{P}\{Z_\alpha \leq y\} dy \right)^{1/2} \text{ a. s.,} \end{aligned}$$

where Z_α is a random variable with Mittag-Leffler distribution with parameter α .



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Lemma 2

The variance of $\hat{N}(t)$ exhibits the following asymptotic behavior, as $t \rightarrow \infty$,

$$\text{Var}[\hat{N}(t)] \sim \left(\frac{\sigma^2 t}{\mu^3 \pi} \right)^{1/2} \quad \text{under (A),}$$

$$\text{Var}[\hat{N}(t)] \sim \left(\frac{1}{\mu^3 \pi} \right)^{1/2} c_2(t) \quad \text{under (B),}$$

$$\text{Var}[\hat{N}(t)] \sim \frac{\Gamma(1 - 1/\alpha)}{\mu^{1+1/\alpha} \pi} \left(\frac{-2\Gamma(2 - \alpha) \cos(\pi\alpha/2)}{\alpha - 1} \right)^{1/\alpha} c_\alpha(t) \quad \text{under (C),}$$

and

$$\text{Var}[\hat{N}(t)] \sim \frac{\int_0^\infty \mathbb{P}\{Z_\alpha > y\} \mathbb{P}\{Z_\alpha \leq y\} dy}{\mathbb{P}\{\xi > t\}} \quad \text{under (D).}$$



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- Apply the LIL for sums of independent indicators, established by [Buraczewski, Iksanov & Kotelnikova \(2025\)](#), to

$$\hat{N}(t) = \sum_{k \geq 1} \mathbf{1}_{\{\hat{S}_k \leq t\}}.$$

- Verify the required conditions – apply lemma 2.
- $\nu := \inf\{\nu^* \geq 1 : \mathbb{E}[\hat{N}(t)] = O((\text{Var}[\hat{N}(t)])^{\nu^*})\}$
 $\nu = 2$ in (A),(B); $\nu = \alpha \in (1, 2)$ in (C); $\nu = 1$ in (D).
- The value of ν determines the type of law:

$$\nu > 1 \Rightarrow \textbf{LSL} \quad (\text{cases (A),(B),(C)});$$

$$\nu = 1 \Rightarrow \textbf{LIL} \quad (\text{case (D)}).$$



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Assume that $\hat{S}_1$ has the gamma distribution with parameters $2/\rho$ and 1. Then $\mu = \sigma^2 = 2/\rho$. Then

$$\limsup_{t \rightarrow \infty} \frac{\Theta_\rho(D_t) - \mathbb{E}\hat{N}(t^\rho)}{t^{\rho/4}(\log t)^{1/2}} = \frac{\rho}{(4\pi)^{1/4}} \quad \text{a. s.,}$$

and

$$\liminf_{t \rightarrow \infty} \frac{\Theta_\rho(D_t) - \mathbb{E}\hat{N}(t^\rho)}{t^{\rho/4}(\log t)^{1/2}} = -\frac{\rho}{(4\pi)^{1/4}} \quad \text{a. s..}$$



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THANK YOU FOR ATTENTION!